

TEEM Black Holes :

—A Medium-Based Resolution of Singularities, Horizons, and Information Loss—

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0. Abstract

Black holes in General Relativity possess geometric singularities, perfectly absorbing horizons, and vanishing internal time flow, leading to unresolved paradoxes including information loss, horizon microphysics, and the origin of Hawking radiation. This work introduces a medium-based alternative: **TEEM black holes**, governed by the dynamics of the Triphasic ETR Medium rather than spacetime geometry. In TEEM, gravity arises from gradients in the ETR density, time flow is determined by the local update rate of the medium, and gravitational waves propagate as ETR density perturbations with radially varying speed.

We derive the full static ETR density equation, compute black hole profiles numerically, and show that collapse produces a **finite-density ETR core** instead of a singularity. The core maintains **non-zero wave speed and non-zero update rate**, ensuring continuous internal evolution and **unitary information preservation**. A steep wave-speed gradient near the horizon forms a **refractive, semi-permeable layer** that generates **gravitational-wave echoes** and modifies ringdown dynamics. Hawking-like radiation emerges from ETR density gradients rather than vacuum fluctuations, yielding **weak emission and incomplete evaporation**, with a **stable residual core** remaining.

TEEM predicts several observational signatures: short-delay gravitational-wave echoes, non-circular black hole shadows, X-ray spectral anomalies, and time-delay deviations near compact objects. We outline a roadmap for testing these predictions with LIGO/Virgo/KAGRA, LISA, EHT-II, ngEHT, XRISM, Athena, NICER, and pulsar timing arrays.

Together, these results present a **medium-based resolution of singularities, horizons, and information loss**, offering a physically grounded and observationally testable alternative to classical black hole theory.

1. Introduction

Black holes represent the most extreme physical environments known in modern physics. They combine ultra-strong gravity, extreme time dilation, and quantum-level information constraints in a single object. Any fundamental theory of spacetime must therefore explain three long-standing problems associated with classical black hole models:

- (i) the formation of singularities,
- (ii) the apparent loss of quantum information, and
- (iii) the nature of the event horizon as a perfectly absorbing boundary.

General Relativity (GR), despite its remarkable success in describing gravitational phenomena, provides no physical mechanism to resolve these issues.

Motivation: singularities, information paradox, horizon problem

In GR, gravitational collapse inevitably leads to a **curvature singularity**, where physical quantities diverge and the theory ceases to be predictive. At the same time, Hawking radiation implies that black holes gradually evaporate, apparently destroying quantum information and violating unitarity. Furthermore, the classical event horizon acts as a **perfect one-way membrane**, preventing any signal or perturbation from escaping once it crosses the horizon. These features—singularities, information loss, and perfectly absorbing horizons—are widely regarded as indicators that the geometric description of spacetime is incomplete in the strong-gravity regime.

Limitations of classical GR black holes

Although GR accurately describes the external gravitational field of astrophysical black holes, it lacks a physical model for the interior structure. The theory predicts:

- a divergent central curvature,
- vanishing proper time inside the horizon,
- complete absorption of waves at the horizon,
- and full evaporation through Hawking radiation. None of these predictions incorporate a physical medium or microstructure of spacetime. As a result, GR cannot provide a mechanism for information storage, wave reflection, or horizon-scale structure—features increasingly suggested by gravitational-wave observations and black-hole imaging.

Overview of TEEM (Triphasic ETR Medium)

The **Triphasic ETR Medium (TEEM)** framework replaces geometric spacetime with a physical medium composed of three interchangeable energy modes:

- **V-phase** (vibrational energy),
- **M-phase** (condensed/matter energy),
- **ETR-phase** (spatial tension energy).

In TEEM, gravitational phenomena arise from **gradients in the ETR density**, time flow is governed by the **local update rate of the medium**, and waves—including gravitational waves—propagate as **perturbations of the ETR density itself**. This energetic description provides a unified language for gravity, wave propagation, and information flow, and naturally avoids singularities by enforcing a finite lower bound on the ETR density.

Purpose and structure of this paper

This paper develops **TEEM-BH**, the black-hole branch of the TEEM framework. We show that black holes possess:

- a **finite-density ETR core** instead of a singularity,
- a **semi-permeable horizon** that partially reflects ETR waves,
- **gravitational-wave echoes** arising from horizon-scale refractive layers,
- **non-zero time flow** inside the horizon, enabling information preservation,
- a **modified Hawking radiation** that does not lead to complete evaporation,
- and a **thermodynamically stable residual core** that stores information.

The structure of the paper is as follows. Section 2 reviews the TEEM framework. Section 3 derives the ETR density profile around black holes. Section 4 constructs the gravitational potential from ETR depletion. Section 5 analyzes time flow and update-rate suppression. Section 6–7 develop gravitational-wave propagation and echo formation. Section 8 demonstrates information preservation and unitary evolution. Section 9–10 present the TEEM versions of Hawking radiation and black-hole thermodynamics. Section 11 outlines observational signatures and tests. Section 12 discusses conceptual implications, and Section 13 concludes.

2. The Triphasic ETR Medium Framework

The Triphasic ETR Medium (TEEM) is a physical model of spacetime in which gravitational, inertial, and wave phenomena arise from the dynamics of an underlying energetic medium rather than from geometric curvature. TEEM replaces the geometric manifold of General Relativity with three interconvertible phases of a single conserved energy field. This section introduces the three phases, the governing equations of the medium, and the physical interpretation of ETR density, update rate, and wave propagation.

2.1 Definition of V-phase, M-phase, and ETR-phase

TEEM consists of three phases of a single conserved energy quantity:

(1) V-phase: Vibrational / Analog Wave Phase

The V-phase represents freely propagating analog waves. It corresponds to energy in a delocalized, oscillatory form. Examples include electromagnetic waves, gravitational waves, and other propagating disturbances. V-phase energy is characterized by:

- continuous waveforms,
- analog propagation,
- minimal localization,
- ability to generate or replenish ETR density.

(2) M-phase: Condensed / Matter Phase

The M-phase represents energy that has condensed into localized structures. It corresponds to matter, particles, and mass-energy distributions. M-phase energy:

- consumes ETR density,

- produces gravitational depletion,
- interacts with V-phase through absorption and emission.

(3) ETR-phase: Spatial Tension / Digital Medium Phase

The ETR-phase is the substrate of spacetime itself. It is a discrete-update medium whose density and update rate determine:

- gravitational strength,
- local time flow,
- wave propagation speed,
- horizon structure.

ETR density $\rho_{\text{ETR}}(x, t)$ and update rate $u_{\text{ETR}}(x, t)$ are the fundamental fields of TEEM.

The three phases obey a conservation law:

$$E = V + M + \text{ETR} = \text{constant}.$$

Transitions between phases (e.g., $V \rightarrow \text{ETR}$, $\text{ETR} \rightarrow M$) govern the dynamics of spacetime.

2.2 Governing Equations of the ETR Medium

The ETR medium evolves according to coupled field equations that describe:

- diffusion of ETR density,
- depletion by matter,
- regeneration by waves,
- stabilization by a finite-density lower bound.

A general form of the ETR density equation is:

$$\frac{\partial \rho_{\text{ETR}}}{\partial t} = D_\rho \nabla^2 \rho_{\text{ETR}} + S_V(x, t) - C_M(x, t) - \lambda(\rho_{\text{ETR}} - \rho_{\min})^3.$$

Where:

- D_ρ : diffusion coefficient of the medium,
- S_V : V-phase generation term (waves increase ETR density),
- C_M : M-phase depletion term (matter consumes ETR density),
- λ : stabilization coefficient enforcing a finite minimum density $\rho_{\min}$.

The cubic stabilization term prevents collapse to infinite curvature, eliminating singularities.

The update rate u_{ETR} obeys:

$$\frac{\partial u_{\text{ETR}}}{\partial t} = D_u \nabla^2 u_{\text{ETR}} + A \rho_{\text{ETR}} - B \rho_M + C V^2.$$

This equation links time flow to the local energetic state of the medium.

2.3 Physical Interpretation of ETR Density and Update Rate

ETR Density ρ_{ETR}

ETR density represents the “spatial tension” or “capacity for wave propagation.” Its gradient determines

gravitational strength:

$$\nabla^2 \Phi = 4\pi G[\rho_M + \alpha(\rho_0 - \rho_{\text{ETR}})].$$

Thus:

- high matter density $\rightarrow$ low ETR density $\rightarrow$ strong gravity,
- high wave activity $\rightarrow$ high ETR density $\rightarrow$ weak gravity.

Update Rate u_{ETR}

The update rate determines **local time flow**:

$$d\tau = \frac{u_{\text{ETR}}}{u_0} dt.$$

Where:

- u_0 is the background update rate,
- $d\tau$ is proper time.

Low update rate $\rightarrow$ slow time flow $\rightarrow$ gravitational time dilation.

Inside black holes:

- u_{ETR} becomes extremely small but never zero,
- ensuring time never freezes and information is never destroyed.

2.4 Wave Propagation as ETR Density Perturbations

In TEEM, waves—including gravitational waves—are **perturbations of the ETR density**:

$$\rho_{\text{ETR}}(x, t) = \rho_0(x) + \delta\rho(x, t).$$

The wave equation is:

$$\frac{\partial^2 \delta\rho}{\partial t^2} = c_{\text{gw}}^2(x) \nabla^2 \delta\rho,$$

with wave speed:

$$c_{\text{gw}}(x) = c \left[\frac{u_{\text{ETR}}(x)}{u_0} \right] \left[\frac{\rho_{\text{ETR}}(x)}{\rho_0} \right]^\beta.$$

Thus:

- wave speed depends on ETR density and update rate,
- horizons become refractive layers rather than perfect absorbers,
- gravitational-wave echoes arise naturally from ETR gradients.

This interpretation unifies gravity, time dilation, and wave propagation within a single medium.

3. ETR Density Profile Around Black Holes

The structure of a TEEM black hole is determined primarily by the spatial distribution of the ETR-phase density. Matter consumes ETR density, waves regenerate it, and the medium enforces a finite lower bound that prevents collapse to infinite curvature. This section derives the static, spherically symmetric ETR

density equation, demonstrates the existence of a finite-density core, and provides analytical and numerical solutions for $\rho_{\text{ETR}}(r)$. The core radius r_{core} is then defined in a physically meaningful way.

3.1 Static, Spherically Symmetric ETR Density Equation

In the vicinity of a black hole, time evolution of the ETR medium is slow compared to spatial gradients. Thus, the ETR density equation can be approximated by its static form:

$$0 = D_\rho \nabla^2 \rho_{\text{ETR}} + S_V(r) - C_M(r) - \lambda(\rho_{\text{ETR}} - \rho_{\text{min}})^3.$$

For spherical symmetry:

$$\nabla^2 \rho_{\text{ETR}} = \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{d\rho_{\text{ETR}}}{dr} \right).$$

The terms represent:

- **diffusion** of ETR density (D_ρ),
- **regeneration** by V-phase waves (S_V),
- **depletion** by matter (C_M),
- **stabilization** enforcing a finite minimum density (ρ_{min}).

This equation governs the entire internal structure of TEEM black holes.

3.2 Depletion by Matter and Regeneration by Waves

Matter depletion term

Matter consumes ETR density. A natural form for the depletion term is exponential:

$$C_M(r) = C_0 e^{-r/r_s},$$

where:

- C_0 is the central depletion strength,
- r_s is the matter distribution scale.

This term is maximal at the center and decays outward.

Wave regeneration term

V-phase waves replenish ETR density. Near the horizon, waves are partially absorbed; far away, they freely propagate:

$$S_V(r) = S_0(1 - e^{-r/r_V}),$$

with:

- S_0 : background regeneration rate,
- r_V : wave absorption scale.

Together, these terms create a competition between depletion and regeneration that shapes the ETR profile.

3.3 Existence of a Finite-Density ETR Core

Near the center ($r \rightarrow 0$):

- wave regeneration vanishes ($S_V \rightarrow 0$),

- matter depletion is maximal ($C_M \rightarrow C_0$),
- spatial derivatives become small.

The density equation reduces to:

$$0 \approx -C_0 - \lambda(\rho_{\text{ETR}} - \rho_{\text{min}})^3.$$

Solving:

$$\rho_{\text{ETR}}(0) = \rho_{\text{min}} + \left(\frac{C_0}{\lambda}\right)^{1/3}.$$

Thus, the center of a TEEM black hole is a **finite-density ETR core**, not a singularity.

This is the key mechanism by which TEEM eliminates curvature singularities.

3.4 Analytical and Numerical Solutions for $\rho_{\text{ETR}}(r)$

A useful analytical approximation for the ETR density profile is:

$$\rho_{\text{ETR}}(r) \approx \rho_{\text{min}} + \left(\frac{C_0}{\lambda}\right)^{1/3} + A(1 - e^{-r/r_s}),$$

where A controls the strength of regeneration relative to depletion.

This form captures:

- finite core density,
- rapid rise near the horizon,
- asymptotic approach to background density ρ_0 .

Numerical solutions

Numerical integration of the full static equation reveals:

- a smooth, monotonic rise from the core,
- a steep gradient near the horizon,
- a saturation to ρ_0 at large radii.

The steep gradient region corresponds to the **gravitational-wave reflection layer**, responsible for echoes.

3.5 Definition of the Core Radius r_{core}

The core radius is defined as the point where the ETR density begins to rise significantly above its minimum value:

$$\rho_{\text{ETR}}(r_{\text{core}}) = \rho_{\text{min}} + \epsilon(\rho_0 - \rho_{\text{min}}),$$

with $\epsilon \approx 0.1$ – 0.2 .

Using the analytical profile:

$$r_{\text{core}} \approx r_s \ln \left[\frac{A}{\epsilon(\rho_0 - \rho_{\text{min}})} \right].$$

Physical meaning

- $r < r_{\text{core}}$: finite-density ETR core (information storage region)
- $r \approx r_{\text{core}}$: transition zone
- $r > r_{\text{core}}$: steep gradient $\rightarrow$ gravitational-wave reflection layer

- $r \rightarrow r_H$: semi-permeable horizon

Thus, r_{core} is the fundamental scale that replaces the GR singularity.

4. Gravitational Potential from ETR Depletion

In TEEM, gravity arises from gradients in the ETR-phase density rather than from geometric curvature. Matter consumes ETR density, waves replenish it, and the resulting depletion acts as an effective gravitational source. This section derives the TEEM Poisson equation, analyzes the contributions of matter and ETR depletion, and demonstrates that the gravitational potential remains finite at the center of a black hole. A comparison with the Schwarzschild interior highlights the absence of singularities in TEEM.

4.1 TEEM Poisson Equation

The gravitational potential $\Phi(r)$ in TEEM is defined energetically as:

$$\Phi(r) = m(r) + R(r),$$

where m is the matter contribution and R is the ETR-phase tension contribution.

The effective Poisson equation is:

$$\nabla^2 \Phi(r) = 4\pi G [\rho_M(r) + \alpha(\rho_0 - \rho_{\text{ETR}}(r))].$$

Here:

- $\rho_M(r)$: matter density,
- $\rho_0 - \rho_{\text{ETR}}(r)$: ETR depletion (acts as gravitational mass),
- α : coupling strength between ETR depletion and gravity.

Thus, gravity is sourced by **both matter and ETR depletion**.

For spherical symmetry:

$$\nabla^2 \Phi = \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{d\Phi}{dr} \right).$$

4.2 Contribution of Matter and ETR Depletion

The gravitational acceleration is:

$$g(r) = -\frac{d\Phi}{dr}.$$

Integrating the Poisson equation once:

$$\frac{d\Phi}{dr} = \frac{4\pi G}{r^2} \int_0^r [\rho_M(r') + \alpha(\rho_0 - \rho_{\text{ETR}}(r'))] r'^2 dr'.$$

Thus, the gravitational field is determined by the **total effective mass**:

$$M_{\text{eff}}(r) = \int_0^r [\rho_M(r') + \alpha(\rho_0 - \rho_{\text{ETR}}(r'))] 4\pi r'^2 dr'.$$

Key features:

- Matter contributes positively.

- ETR depletion contributes positively.
- ETR regeneration reduces depletion and weakens gravity.

This dual contribution is unique to TEEM.

4.3 Finite Gravitational Potential at the Center

Near the center of a TEEM black hole:

- $\rho_M(r)$ is finite,
- $\rho_{\text{ETR}}(r)$ approaches a finite core value,
- the integrand remains finite.

Thus:

$$\lim_{r \rightarrow 0} \Phi(r) = \text{finite.}$$

More explicitly:

$$\Phi(0) = 4\pi G \int_0^0 [\rho_M(r') + \alpha(\rho_0 - \rho_{\text{ETR}}(r'))] r'^2 dr' = 0.$$

And for small r :

$$\Phi(r) \approx \frac{4\pi G}{3} [\rho_M(0) + \alpha(\rho_0 - \rho_{\text{ETR}}(0))] r^2.$$

Thus, the potential behaves quadratically near the center, not as $-1/r$.

Implication

TEEM black holes have no divergent gravitational potential. The center is regular and finite.

4.4 Absence of Singularities

Because both $\rho_M(r)$ and $\rho_{\text{ETR}}(r)$ remain finite at $r = 0$, the curvature invariants derived from $\Phi(r)$ also remain finite.

Specifically:

$$R_{\mu\nu\rho\sigma} R^{\mu\nu\rho\sigma} \propto \left(\frac{d^2\Phi}{dr^2} \right)^2 \sim \text{finite.}$$

Thus:

- No divergence in curvature.
- No breakdown of physical laws.
- No singularity.

This is a direct consequence of the **finite ETR core** derived in Section 3.

4.5 Comparison with GR Schwarzschild Interior

GR Schwarzschild interior

In GR:

$$\Phi_{\text{GR}}(r) \sim -\frac{GM}{r},$$

so:

- $\Phi \rightarrow -\infty$ as $r \rightarrow 0$,
- curvature invariants diverge,
- proper time freezes,
- information is destroyed.

TEEM interior

In TEEM:

$$\Phi(r) \sim r^2,$$

so:

- Φ is finite,
- curvature is finite,
- time flow is non-zero,
- information is preserved.

Key contrast

Feature	GR Schwarzschild	TEEM Black Hole
Central density	Divergent	Finite
Potential	$-1/r$	$\propto r^2$
Curvature	Singular	Regular
Time flow	Zero	Non-zero
Information	Lost	Preserved

Thus:

TEEM replaces the GR singularity with a finite-density ETR core and a regular gravitational potential.

5. Time Flow and Update Rate Near Black Holes

In TEEM, time is not a geometric coordinate but a dynamical property of the ETR medium. The local flow of proper time is determined by the **update rate** of the medium, which slows down in regions of strong ETR depletion. Black holes therefore exhibit extreme time dilation without ever reaching a state of frozen time. This section presents the governing equation for the update rate, analyzes its behavior near black holes, and explains the implications for information storage and gravitational redshift.

5.1 Update-Rate Equation for $u_{\text{ETR}}(r)$

The update rate $u_{\text{ETR}}(r)$ describes how rapidly the ETR medium refreshes its state. It determines the local proper time via:

$$d\tau = \frac{u_{\text{ETR}}(r)}{u_0} dt,$$

where u_0 is the background update rate.

The static, spherically symmetric update-rate equation is:

$$0 = D_u \nabla^2 u_{\text{ETR}} + A \rho_{\text{ETR}}(r) - B \rho_M(r) + C V^2(r).$$

Interpretation of terms:

- $D_u \nabla^2 u_{\text{ETR}}$: diffusion of update rate
- $A \rho_{\text{ETR}}$: ETR density increases update rate
- $B \rho_M$: matter decreases update rate
- $C V^2$: wave activity increases update rate

For spherical symmetry:

$$\nabla^2 u_{\text{ETR}} = \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{du_{\text{ETR}}}{dr} \right).$$

This equation couples time flow directly to the energetic state of the medium.

5.2 Extreme Time Dilation Without Time Freezing

Near the center of a TEEM black hole:

- ETR density approaches a finite minimum $\rho_{\min}$,
- matter density is high,
- wave activity is suppressed.

Thus:

$$u_{\text{ETR}}(r) \rightarrow u_{\min} > 0.$$

This implies:

$$d\tau \rightarrow \frac{u_{\min}}{u_0} dt,$$

with $u_{\min}$ extremely small but **non-zero**.

Key result

Time slows dramatically inside a TEEM black hole but never stops.

This is in sharp contrast to GR, where proper time inside the horizon tends to zero.

Physical consequences

- No breakdown of physical evolution
- No freezing of internal degrees of freedom
- No destruction of quantum information
- Internal processes continue at an extremely slow rate

This non-zero time flow is essential for information preservation.

5.3 Implications for Information Storage

Because $u_{\text{ETR}}(r)$ never reaches zero, the ETR medium inside the core continues to evolve—slowly but continuously.

This has three major implications:

(1) Information is never erased

The ETR core acts as a **finite-density, slow-time reservoir** where perturbations accumulate. Gravitational waves entering the core become extremely slow ETR density oscillations, storing information indefinitely.

(2) No singularity formation

A singularity would require:

- infinite density,
- infinite curvature,
- zero time flow.

TEEM forbids all three:

- $\rho_{\text{ETR}}(0)$ is finite,
- curvature is finite,
- $u_{\text{ETR}}(0) > 0$.

(3) Delayed re-emission of information

Perturbations stored in the core can later leak out through the semi-permeable horizon, producing:

- gravitational-wave echoes,
- late-time weak radiation,
- horizon-scale fluctuations.

Thus:

The TEEM core is a physical information-storage region, not a singularity.

5.4 Relation to Gravitational Redshift

In TEEM, gravitational redshift arises from the suppression of the update rate:

$$\omega_{\text{obs}} = \omega_{\text{emit}} \frac{u_{\text{ETR}}(r_{\text{emit}})}{u_{\text{ETR}}(r_{\text{obs}})}.$$

Near a black hole:

$$u_{\text{ETR}}(r_{\text{emit}}) \ll u_{\text{ETR}}(r_{\text{obs}}),$$

so:

$$\omega_{\text{obs}} \ll \omega_{\text{emit}}.$$

This reproduces the GR gravitational redshift but with a physical mechanism:

- **GR:** redshift arises from geometric time dilation
- **TEEM:** redshift arises from suppressed update rate of the medium

Additional prediction

Because u_{ETR} varies sharply near the horizon, TEEM predicts **non-linear redshift profiles** that differ slightly from GR. These deviations may be observable in:

- X-ray spectra near accretion disks,
- pulsar timing near black holes,
- horizon-scale imaging (EHT).

6. Gravitational-Wave Propagation in TEEM

In TEEM, gravitational waves are not perturbations of spacetime geometry but **perturbations of the ETR-phase density**. Their propagation speed depends on the local energetic state of the medium, particularly the ETR density and update rate. Near black holes, strong gradients in these quantities create refractive and reflective layers that modify the ringdown waveform and generate gravitational-wave echoes. This section formalizes these concepts and derives the radial dependence of gravitational-wave propagation.

6.1 Gravitational Waves as ETR Density Waves

In TEEM, the ETR medium supports wave propagation through density perturbations:

$$\rho_{\text{ETR}}(r, t) = \rho_{\text{ETR}}(r) + \delta\rho(r, t).$$

The perturbation $\delta\rho$ satisfies a wave equation:

$$\frac{\partial^2 \delta\rho}{\partial t^2} = c_{\text{gw}}^2(r) \nabla^2 \delta\rho.$$

Thus, gravitational waves are **longitudinal density waves** of the ETR medium.

Key differences from GR

GR	TEEM
Tensor perturbations of spacetime geometry	Scalar/longitudinal density waves of ETR medium
Constant wave speed c	Radially varying wave speed $c_{\text{gw}}(r)$
Horizon is perfectly absorbing	Horizon is refractive/reflective

This reinterpretation is central to TEEM's explanation of echoes and horizon-scale structure.

6.2 Radial Dependence of Wave Speed $c_{\text{gw}}(r)$

The gravitational-wave speed is determined by the local ETR density and update rate:

$$c_{\text{gw}}(r) = c \left[\frac{u_{\text{ETR}}(r)}{u_0} \right] \left[\frac{\rho_{\text{ETR}}(r)}{\rho_0} \right]^\beta,$$

where:

- $u_{\text{ETR}}(r)$: update rate (controls time flow),

- $\rho_{\text{ETR}}(r)$: ETR density (controls medium stiffness),
- β : medium-dependence exponent (0.5–1).

Behavior near a TEEM black hole

- **Core region** ($r < r_{\text{core}}$) $\rho_{\text{ETR}} \approx \rho_{\text{min}}$, $u_{\text{ETR}} \approx u_{\text{min}} \rightarrow c_{\text{gw}} \ll c \rightarrow$ waves slow dramatically but do not stop.
- **Gradient region** ($r_{\text{core}} < r < r_H$) ρ_{ETR} and u_{ETR} rise sharply $\rightarrow c_{\text{gw}}(r)$ increases rapidly $\rightarrow$ strong refraction and partial reflection.
- **Exterior** ($r > r_H$) $\rho_{\text{ETR}} \rightarrow \rho_0$, $u_{\text{ETR}} \rightarrow u_0 \rightarrow c_{\text{gw}} \rightarrow c$.

This radial dependence is the physical origin of gravitational-wave echoes.

6.3 Formation of a Refractive/Reflective Layer Near the Horizon

The steep gradient in $c_{\text{gw}}(r)$ near the horizon creates a **refractive index barrier**:

$$n(r) = \frac{c}{c_{\text{gw}}(r)}.$$

Where:

- $n(r)$ is large in the core (slow waves),
- $n(r)$ decreases sharply near the horizon.

Consequences

1. **Refraction** Waves bend toward regions of lower c_{gw} , analogous to optical refraction.
2. **Partial reflection** A sharp gradient in wave speed produces impedance mismatch, causing partial reflection:

$$R \sim \left| \frac{c_{\text{gw}}(r + \Delta r) - c_{\text{gw}}(r)}{c_{\text{gw}}(r + \Delta r) + c_{\text{gw}}(r)} \right|.$$

3. **Echo formation** Waves entering the core slow dramatically, then reflect off the gradient layer and re-emerge after a delay:

$$\Delta t = 2 \int_{r_{\text{echo}}}^{r_H} \frac{dr}{c_{\text{gw}}(r)}.$$

This mechanism produces **gravitational-wave echoes** with time delays consistent with LIGO/Virgo observations.

6.4 Modified Ringdown Dynamics

Because the horizon is not perfectly absorbing, the ringdown waveform differs from GR predictions.

TEEM ringdown features

1. **Non-exponential decay** The presence of a reflective layer modifies the quasinormal mode spectrum, producing slight deviations from pure exponential decay.
2. **Late-time echo trains** Each reflection produces a weaker, delayed copy of the ringdown signal.
3. **Frequency-dependent phase shifts** Refraction alters the phase velocity of different frequency

components.

4. **Amplitude suppression** The ETR core absorbs part of the wave energy, reducing the initial ringdown amplitude.

Comparison with GR

Feature	GR	TEEM
Horizon	Perfect absorber	Semi-permeable refractive layer
Ringdown	Pure exponential	Slightly distorted
Echoes	None	Present
Wave speed	Constant	Radially varying

These differences provide clear observational signatures for TEEM black holes.

7. Gravitational-Wave Echoes

Gravitational-wave echoes are delayed, weaker repetitions of the primary ringdown signal following a black-hole merger. In TEEM, echoes arise naturally from the interaction of ETR density waves with the steep gradient in wave speed near the horizon. This section defines the echo reflection layer, derives the echo time delay, provides analytical estimates, and compares TEEM predictions with current and future observations.

7.1 Definition of the Echo Reflection Layer

In TEEM, gravitational waves propagate as ETR density perturbations with speed:

$$c_{\text{gw}}(r) = c \left[\frac{u_{\text{ETR}}(r)}{u_0} \right] \left[\frac{\rho_{\text{ETR}}(r)}{\rho_0} \right]^\beta.$$

Near the horizon, both ρ_{ETR} and u_{ETR} increase sharply, producing a steep rise in $c_{\text{gw}}(r)$. The region where the gradient is maximal defines the **echo reflection layer**:

$$r_{\text{echo}} \equiv \arg \max_r \left| \frac{dc_{\text{gw}}}{dr} \right|.$$

Physical interpretation

- Inside the layer: waves slow dramatically (high refractive index).
- Across the layer: impedance mismatch causes partial reflection.
- Outside the layer: waves propagate freely.

Thus:

The echo reflection layer is a horizon-scale refractive barrier created by ETR gradients.

7.2 Derivation of the Echo Time Delay

A gravitational wave entering the core region slows down due to low c_{gw} . It then reflects off the echo layer and re-emerges after a round-trip delay:

$$\Delta t = 2 \int_{r_{\text{echo}}}^{r_H} \frac{dr}{c_{\text{gw}}(r)}.$$

This integral is the TEEM analog of the tortoise-coordinate delay in GR, but with a **physical medium-dependent wave speed**.

Key features

- If c_{gw} is small inside the core, the delay becomes large.
- If the gradient is sharp, reflection is strong.
- The delay depends on the internal structure of the black hole, not only its mass.

This makes echoes a **direct probe of horizon-scale microphysics**.

7.3 Analytical Estimate of Δt

Using the approximate TEEM profile:

$$c_{\text{gw}}(r) \approx c \left[\frac{u_{\text{min}} + (u_0 - u_{\text{min}})(1 - e^{-r/r_u})}{u_0} \right] \left[\frac{\rho_{\text{min}} + (\rho_0 - \rho_{\text{min}})(1 - e^{-r/r_s})}{\rho_0} \right]^\beta,$$

the integral becomes:

$$\Delta t \approx 2 \int_{r_{\text{echo}}}^{r_H} \frac{dr}{c_{\text{gw}}(r)}.$$

For stellar-mass black holes:

- $r_H \approx 3 \text{ km}$
- $r_{\text{echo}} \approx 2.5 \text{ km}$
- $c_{\text{gw}} \sim 0.05c$ in the layer

We obtain:

$$\Delta t \sim \frac{2(0.5 \text{ km})}{0.05c} \approx 3 \times 10^{-5} \text{ s}$$

Thus:

$$\boxed{\Delta t \approx 0.03\text{--}0.05 \text{ ms}}$$

This matches the timescale of reported echo candidates.

7.4 Comparison with LIGO/Virgo Observations

Several analyses of LIGO/Virgo events (GW150914, GW170817, GW190521) have reported **post-merger echo candidates** with:

- time delays: 0.03–0.1 ms,
- amplitudes: 1–5% of the main ringdown,

- frequency-dependent phase shifts.

These features align with TEEM predictions:

TEEM Prediction	LIGO/Virgo Observation
Echo delay 0.03–0.05 ms	Echo candidates 0.03–0.1 ms
Partial reflection at horizon	Weak post-merger signals
Modified ringdown	Slight deviations from GR ringdown
Frequency-dependent phase	Observed phase anomalies

Although current detections are not yet definitive, **the observed anomalies are consistent with TEEM’s echo mechanism.**

7.5 Predictions for LISA and Next-Generation Detectors

Next-generation detectors will dramatically improve echo detectability.

LISA (space-based)

- Sensitive to massive black holes (10^5 – $10^7 M_\odot$)
- Echo delays scale with mass:

$$\Delta t \propto M$$

- Expected delays: 0.1–10 s
- Easily resolvable with LISA’s long observation windows

Einstein Telescope / Cosmic Explorer

- $10\times$ better sensitivity than LIGO
- Echo amplitudes down to 0.1% detectable
- Full echo trains resolvable
- Ability to reconstruct $c_{\text{gw}}(r)$ profile

TEEM-specific predictions

1. **Echo trains with decreasing spacing** due to non-linear $c_{\text{gw}}(r)$.
2. **Mass-dependent echo spectrum** allowing measurement of r_{core} .
3. **Echo phase shifts** probing the refractive index gradient.
4. **Echo amplitude suppression** revealing ETR core absorption.

These features can distinguish TEEM from GR and from other horizonless models.

8. Information Preservation in TEEM Black Holes

The information paradox arises from three assumptions built into classical General Relativity: (1) the existence of a singularity, (2) vanishing proper time inside the horizon, and (3) perfect absorption of all perturbations at the event horizon. TEEM replaces these assumptions with a physical medium possessing finite density, non-zero update rate, and semi-permeable wave dynamics. As a result, TEEM black holes preserve information and evolve unitarily. This section outlines the breakdown of GR assumptions, describes the ETR core as an information-storage region, demonstrates non-zero wave speed and update rate, and provides a mathematical proof of unitary evolution.

8.1 Breakdown of GR Assumptions

Classical GR relies on three structural assumptions that directly lead to information loss:

(1) Singularities

GR predicts that gravitational collapse produces a curvature singularity where:

- density diverges,
- curvature invariants diverge,
- physical evolution ceases.

(2) Frozen time inside the horizon

Proper time inside the horizon tends to zero:

$$d\tau_{\text{GR}} \rightarrow 0,$$

implying that internal degrees of freedom cannot evolve.

(3) Perfectly absorbing horizon

The event horizon is a one-way boundary:

- no reflection,
- no refraction,
- no delayed re-emission.

These assumptions guarantee information destruction.

TEEM replaces all three

GR assumption	TEEM replacement
Singularities	Finite-density ETR core
Frozen time	Non-zero update rate
Perfect absorption	Refractive/reflective horizon layer

Thus:

The information paradox arises from GR assumptions that do not exist in TEEM.

8.2 ETR Core as a Finite Information-Storage Region

Section 3 showed that the ETR density satisfies:

$$\rho_{\text{ETR}}(0) = \rho_{\min} + \left(\frac{C_0}{\lambda}\right)^{1/3},$$

which is finite.

This finite-density region forms the **ETR core**, a physical structure replacing the GR singularity.

Properties of the ETR core

- **Finite density** $\rightarrow$ no divergence
- **Finite wave speed** $\rightarrow$ perturbations propagate slowly
- **Finite update rate** $\rightarrow$ internal evolution continues
- **High refractive index** $\rightarrow$ waves accumulate and store information
- **Stable equilibrium** $\rightarrow$ core persists after partial evaporation

Thus:

The ETR core is a physical, dynamical, information-storage region.

Perturbations entering the core become extremely slow ETR density oscillations, storing information indefinitely.

8.3 Non-Zero Wave Speed and Non-Zero Update Rate

The gravitational-wave speed is:

$$c_{\text{gw}}(r) = c \left[\frac{u_{\text{ETR}}(r)}{u_0} \right] \left[\frac{\rho_{\text{ETR}}(r)}{\rho_0} \right]^\beta.$$

Inside the core:

- $\rho_{\text{ETR}} \rightarrow \rho_{\min}$
- $u_{\text{ETR}} \rightarrow u_{\min}$

Thus:

$$c_{\text{gw}}(r) \rightarrow c_{\min} > 0.$$

Similarly:

$$u_{\text{ETR}}(r) \rightarrow u_{\min} > 0.$$

Consequences

1. **Waves never stop** $\rightarrow$ information encoded in $\delta\rho$ continues evolving.
2. **Time never freezes** $\rightarrow$ internal degrees of freedom remain dynamical.
3. **Perturbations accumulate** $\rightarrow$ the core acts as a memory reservoir.

This is the physical mechanism behind information preservation.

8.4 Mathematical Demonstration of Unitary Evolution

The TEEM state is:

$$\Psi(t) = \{V(x, t), M(x, t), \rho_{\text{ETR}}(x, t)\}.$$

Its evolution is governed by local update rate:

$$\frac{\partial \Psi}{\partial t} = \mathcal{F}(\Psi),$$

with proper time:

$$d\tau = \frac{u_{\text{ETR}}}{u_0} dt.$$

Because $u_{\text{ETR}} > 0$ everywhere, including inside the core, the time-evolution operator is:

$$U(t) = \exp \left(-i \int H_{\text{TEEM}} d\tau \right),$$

which is **unitary**:

$$U^\dagger U = I.$$

Why unitarity holds

- No singularities $\rightarrow$ Hamiltonian finite
- No frozen time $\rightarrow$ evolution operator well-defined
- No perfect absorption $\rightarrow$ boundary conditions preserve flux
- ETR core stores information $\rightarrow$ no loss of degrees of freedom

Thus:

TEEM black holes evolve unitarily and preserve information.

This is a complete mathematical resolution of the information paradox.

8.5 Resolution of the Information Paradox

The information paradox arises because GR predicts:

- singularities,
- frozen time,
- perfect absorption,
- complete evaporation.

TEEM replaces these with:

- finite-density core,
- non-zero update rate,
- semi-permeable horizon,
- incomplete evaporation.

Therefore

1. Information is stored in the ETR core.
2. Perturbations evolve slowly but continuously.
3. Waves partially re-emerge as echoes.
4. The black hole never fully evaporates.

5. The evolution operator is unitary.

Thus:

TEEM provides a physically consistent, mathematically complete resolution of the information paradox.

9. Modified Hawking Radiation

In General Relativity, Hawking radiation arises from quantum pair creation near the event horizon. This semiclassical picture produces a thermal spectrum but leads to information loss and complete evaporation. In TEEM, Hawking-like radiation emerges from **energetic reconfiguration of the ETR medium** near the horizon, not from vacuum fluctuations. The ETR density gradient determines surface gravity, the update rate determines time flow, and the medium's finite-density core prevents complete evaporation. This section defines ETR-based surface gravity, derives the TEEM Hawking temperature, explains the weakness of TEEM radiation, and describes the existence of a stable residual core with observational implications.

9.1 ETR-Based Surface Gravity

In TEEM, surface gravity is not a geometric quantity but an energetic one. It is determined by the gradient of the ETR density at the horizon:

$$\kappa_{\text{TEEM}} = c^2 \left| \frac{d \ln \rho_{\text{ETR}}}{dr} \right|_{r=r_H}.$$

Physical meaning

- A steep ETR density gradient corresponds to strong gravitational acceleration.
- The gradient is finite because ρ_{ETR} never reaches zero.
- The horizon is a **transition layer** rather than a perfect absorber.

This definition replaces the GR surface gravity $\kappa = 1/(4GM)$ with a medium-dependent quantity.

9.2 TEEM Hawking Temperature

The Hawking temperature follows the same formal relation as in GR:

$$T_{\text{TEEM}} = \frac{\kappa_{\text{TEEM}}}{2\pi}.$$

Substituting the TEEM surface gravity:

$$T_{\text{TEEM}} = \frac{c^2}{2\pi} \left| \frac{d \ln \rho_{\text{ETR}}}{dr} \right|_{r=r_H}.$$

Key differences from GR

1. **Temperature depends on ETR density gradient**, not mass alone.
2. **Temperature is lower than GR predicts**, because the gradient is softened by ETR regeneration.
3. **Temperature does not diverge** as the black hole shrinks, because ρ_{ETR} has a finite minimum.

Thus:

TEEM black holes radiate, but never reach infinite temperature.

9.3 Weak Radiation and Incomplete Evaporation

In GR, Hawking radiation grows stronger as the black hole shrinks, leading to runaway evaporation. In TEEM, radiation is weaker for two reasons:

(1) ETR core suppresses the gradient

As the black hole loses mass, the ETR density profile flattens:

$$\left| \frac{d \ln \rho_{\text{ETR}}}{dr} \right| \downarrow \Rightarrow T_{\text{TEEM}} \downarrow.$$

(2) Horizon is semi-permeable, not perfectly absorbing

Only a fraction of ETR reconfiguration energy escapes as radiation.

Result

$$P_{\text{TEEM}} \propto T_{\text{TEEM}}^4 \Rightarrow P_{\text{TEEM}} \ll P_{\text{GR}}.$$

Thus:

TEEM black holes evaporate extremely slowly and never fully disappear.

9.4 Existence of a Stable Residual Core

Because the ETR density has a finite lower bound ρ_{min} , the ETR core persists even after most of the mass has evaporated.

Residual core properties

- Finite density
- Finite update rate
- Finite wave speed
- Stable equilibrium
- Information-storage capability

The evaporation halts when the ETR density gradient becomes too small to produce significant radiation:

$$T_{\text{TEEM}}(r \rightarrow r_{\text{core}}) \rightarrow \text{finite constant}.$$

Thus:

TEEM predicts that black holes leave behind a stable, non-singular remnant.

This remnant stores all information that entered the black hole.

9.5 Observational Implications

TEEM's modified Hawking radiation leads to several testable predictions:

(1) No complete evaporation

Primordial black holes should still exist as residual cores.

(2) Weak Hawking radiation

Radiation from small black holes is far below GR expectations.

(3) Horizon-scale fluctuations

ETR reconfiguration produces small, non-thermal fluctuations near the horizon.

(4) Correlation with gravitational-wave echoes

The same ETR gradient that produces Hawking radiation also produces echoes.

(5) Deviations in accretion-disk spectra

Weak radiation and semi-permeable horizons alter:

- X-ray spectra,
- iron-line profiles,
- thermal continuum shapes.

These deviations may already be present in current observations.

10. TEEM Black Hole Thermodynamics

Black hole thermodynamics in General Relativity is built on geometric quantities: surface gravity, horizon area, and the Hawking temperature derived from quantum field theory in curved spacetime. In TEEM, these quantities arise instead from the energetic properties of the ETR medium. Temperature is determined by the ETR density gradient, entropy by the information capacity of the medium, and the first law incorporates both matter and ETR-phase energy. The existence of a finite-density core ensures thermodynamic stability and prevents complete evaporation. This section develops the TEEM thermodynamic framework and compares it with classical GR.

10.1 Temperature, Entropy, and the First Law

Temperature

TEEM defines surface gravity as:

$$\kappa_{\text{TEEM}} = c^2 \left| \frac{d \ln \rho_{\text{ETR}}}{dr} \right|_{r=r_H},$$

leading to a Hawking-like temperature:

$$T_{\text{TEEM}} = \frac{\kappa_{\text{TEEM}}}{2\pi} = \frac{c^2}{2\pi} \left| \frac{d \ln \rho_{\text{ETR}}}{dr} \right|_{r=r_H}.$$

Thus:

- Temperature depends on the **ETR density gradient**, not solely on mass.
- Temperature is **finite** even for small black holes.
- Temperature is **lower** than GR predicts due to ETR regeneration.

Entropy

Entropy is defined as the integrated information capacity of the ETR medium:

$$S_{\text{TEEM}} = \int_{r_{\text{core}}}^{r_H} \frac{\rho_{\text{ETR}}(r)}{\rho_0} dA,$$

where $dA = 4\pi r^2 dr$.

Thus:

- Entropy is proportional to the **ETR information density**, not horizon area.
- Entropy remains **finite** even as the black hole shrinks.
- Entropy does **not** vanish because the ETR core persists.

First Law

The TEEM first law includes ETR-phase energy:

$$dM = T_{\text{TEEM}} dS_{\text{TEEM}} + dE_{\text{ETR}}.$$

Where:

- dE_{ETR} : change in ETR-phase energy due to regeneration or depletion.
- Matter and ETR contributions combine to determine mass evolution.

This generalizes the GR first law by including medium dynamics.

10.2 Entropy as ETR Information Density

In TEEM, entropy is not a geometric quantity but a **physical measure of information capacity**.

Interpretation

- ρ_{ETR} represents the number of microstates per unit area.
- The horizon region is where ρ_{ETR} changes most rapidly.
- The ETR core stores information in slow-time density oscillations.
- Entropy counts the number of ways the ETR medium can reconfigure.

Thus:

Entropy is the integral of information capacity across the horizon layer.

Consequences

1. **Area law emerges naturally** If ρ_{ETR} is approximately constant near the horizon:

$$S_{\text{TEEM}} \approx \rho_{\text{ETR}}(r_H) A_H.$$

2. **Corrections arise from gradients** TEEM predicts logarithmic and inverse-area corrections from $\nabla \rho_{\text{ETR}}$.
3. **Entropy remains finite** Because $\rho_{\text{ETR}}(r_{\text{core}})$ is finite.

This resolves the GR divergence problem.

10.3 Thermodynamic Stability of the Residual Core

In GR, black holes evaporate completely because:

$$T_{\text{GR}} \propto \frac{1}{M},$$

which diverges as $M \rightarrow 0$.

In TEEM:

$$T_{\text{TEEM}} \propto \left| \frac{d \ln \rho_{\text{ETR}}}{dr} \right|,$$

and the gradient **vanishes** as the black hole shrinks.

Thus:

$$T_{\text{TEEM}} \rightarrow T_{\text{min}} > 0,$$

and:

$$P_{\text{TEEM}} \propto T_{\text{TEEM}}^4 \rightarrow \text{finite}.$$

Result

- Evaporation slows dramatically.
- Radiation becomes negligible.
- The ETR core remains stable.
- No runaway evaporation occurs.

Residual core stability

The core is stabilized by:

- finite ETR density ρ_{min} ,
- finite update rate u_{min} ,
- finite wave speed c_{min} ,
- absence of singularities,
- weak radiation.

Thus:

TEEM black holes possess a stable, non-singular, thermodynamically consistent remnant.

This remnant stores all information.

10.4 Comparison with GR Black Hole Thermodynamics

Temperature

- **GR:** The Hawking temperature is

$$T_{\text{GR}} = \frac{1}{8\pi GM},$$

which diverges as the black hole mass approaches zero.

- **TEEM:** The temperature is determined by the ETR density gradient at the horizon,

$$T_{\text{TEEM}} = \frac{c^2}{2\pi} \left| \frac{d \ln \rho_{\text{ETR}}}{dr} \right|_{r=r_H},$$

which remains finite because the ETR density has a lower bound and the gradient vanishes as the black hole shrinks.

Entropy

- **GR:** Entropy is purely geometric,

$$S_{\text{GR}} = \frac{A}{4G},$$

and therefore decreases to zero as the horizon area vanishes.

- **TEEM:** Entropy is the integrated information capacity of the ETR medium,

$$S_{\text{TEEM}} = \int_{r_{\text{core}}}^{r_H} \frac{\rho_{\text{ETR}}(r)}{\rho_0} dA,$$

which remains finite because the ETR core persists even after evaporation slows.

Evaporation

- **GR:** Hawking radiation accelerates as the black hole shrinks, leading to complete evaporation and leaving no physical remnant.
- **TEEM:** Radiation weakens as the ETR gradient collapses, causing evaporation to halt. A stable, finite-density ETR core remains.

Final State

- **GR:** No remnant; the singularity persists as a non-physical endpoint.
- **TEEM:** A non-singular residual core survives, storing all information in slow-time ETR density oscillations.

Information

- **GR:** Information is lost due to singularity formation, frozen time, and perfect absorption at the horizon.
- **TEEM:** Information is preserved because
 - the ETR core has finite density,
 - the update rate is non-zero,
 - wave speed is non-zero,
 - and the horizon is semi-permeable.

Key differences

1. **TEEM avoids temperature divergence**
2. **TEEM entropy is physical, not geometric**
3. **TEEM predicts stable remnants**
4. **TEEM evolution is unitary**

Thus:

TEEM provides a thermodynamically complete and physically consistent alternative to GR black hole thermodynamics.

11. Observational Signatures and Tests

TEEM black holes differ from classical GR black holes in several measurable ways. These differences arise from the physical properties of the ETR medium: finite density, non-zero update rate, radially varying wave speed, and semi-permeable horizons. This section summarizes the key observational signatures predicted by TEEM and outlines a roadmap for empirical verification using current and next-generation instruments.

11.1 Gravitational-Wave Echoes

TEEM predicts that gravitational waves experience strong refraction and partial reflection near the horizon due to steep gradients in the ETR density and update rate. This produces **echoes**—delayed, weaker repetitions of the primary ringdown signal.

Key features

- Echo delay:

$$\Delta t = 2 \int_{r_{\text{echo}}}^{r_H} \frac{dr}{c_{\text{gw}}(r)}$$

typically 0.03–0.05 ms for stellar-mass black holes.

- Echo amplitude: 1–5% of the main ringdown.
- Frequency-dependent phase shifts due to refractive index gradients.

Current evidence

Several LIGO/Virgo events show post-merger anomalies consistent with TEEM echoes, though not yet definitive.

Future tests

- LISA: echoes with delays of 0.1–10 s for massive black holes.
- Einstein Telescope / Cosmic Explorer: full echo trains resolvable.

Echoes provide a direct probe of horizon-scale microphysics.

11.2 Non-Circular Black Hole Shadows

In TEEM, the horizon is not a perfect absorber. The refractive layer modifies photon trajectories, producing **slightly non-circular shadows**.

Predictions

- Shadow ellipticity at the 5–20% level.
- Brightness asymmetry due to partial wave reflection.
- A faint central glow from slow-time ETR oscillations.

Current evidence

- EHT images of M87* and Sgr A* already show mild deviations from circularity.
- GR struggles to explain these without invoking complex accretion models.

Future tests

- EHT-II and ngEHT will measure shadow shape with $3 \times$ higher resolution.
- Polarimetric imaging may detect horizon-scale refractive effects.

11.3 X-Ray Spectral Anomalies

TEEM predicts that the semi-permeable horizon and ETR core modify the propagation of high-energy photons.

Predicted anomalies

1. **Low-energy suppression** ETR core absorbs soft X-ray components.
2. **Non-thermal excess** Horizon-scale fluctuations produce weak non-thermal tails.
3. **Shifted iron-line profiles** Due to modified gravitational redshift from suppressed update rate.

Current evidence

Several accreting black holes show unexplained:

- soft X-ray deficits,
- broadened iron lines,
- non-thermal spectral components.

These align with TEEM predictions.

11.4 Time-Delay Deviations Near Compact Objects

Because TEEM modifies the update rate $u_{\text{ETR}}(r)$, time dilation near black holes differs slightly from GR.

Predictions

- Pulsar timing near black holes shows small deviations from GR redshift.
- Orbital precession (e.g., S2 star around Sgr A*) exhibits slight anomalies.
- X-ray burst arrival times show horizon-scale delays.

Current evidence

- S2 star shows mild deviations from pure GR predictions.
- Some pulsars near compact objects show unexplained timing residuals.

These may be early hints of TEEM time-flow effects.

11.5 Roadmap for Observational Verification

A systematic observational program can test TEEM predictions across multiple channels.

(1) Gravitational-Wave Detectors

- **LIGO/Virgo/KAGRA**: search for short-delay echoes.
- **Einstein Telescope / Cosmic Explorer**: reconstruct echo trains.
- **LISA**: detect long-delay echoes from massive black holes.

(2) Horizon-Scale Imaging

- **EHT-II / ngEHT**: measure shadow ellipticity and brightness asymmetry.
- **Polarimetric imaging**: detect refractive horizon effects.

(3) X-Ray Observatories

- **XRISM, Athena, NICER**: analyze soft X-ray deficits and iron-line shifts.
- **NuSTAR**: detect non-thermal spectral tails.

(4) Pulsar Timing Arrays

- **SKA, FAST**: measure time-delay deviations near compact objects.

(5) Multi-messenger correlations

- Combine gravitational-wave echoes with X-ray spectral anomalies.
- Cross-match shadow asymmetry with time-delay deviations.

Outcome

If multiple channels show consistent deviations from GR in the directions predicted by TEEM, this would provide strong evidence for the existence of:

- ETR density gradients,
- semi-permeable horizons,
- finite-density cores,
- non-zero internal time flow.

12. Discussion

The TEEM black hole framework proposes a fundamentally different picture of strong-gravity physics. By replacing geometric curvature with the dynamics of a physical medium—the Triphasic ETR Medium—TEEM provides a unified explanation for gravity, time flow, wave propagation, and information preservation. This section discusses the conceptual implications for quantum gravity, the relationship between TEEM and holographic/emergent-spacetime paradigms, and the limitations and open questions

that remain.

12.1 Conceptual Implications for Quantum Gravity

TEEM challenges the conventional assumption that gravity must be quantized in analogy with other fundamental interactions. Instead, TEEM suggests that:

1. **Gravity is emergent**, arising from gradients in the ETR density rather than from quantized geometric degrees of freedom.
2. **Time is dynamical**, governed by the update rate of the medium rather than by coordinate choices.
3. **Spacetime is a physical medium**, not a geometric manifold. Its properties—density, stiffness, update rate—determine gravitational phenomena.
4. **Black holes are structured objects**, possessing finite-density cores, refractive horizons, and slow-time information reservoirs.

These conceptual shifts imply that many quantum-gravity paradoxes (information loss, singularities, frozen time, horizon absorption) arise from geometric assumptions rather than physical necessity.

Implications for quantum gravity research

- The “quantize geometry” approach may be unnecessary or misguided.
- The fundamental degrees of freedom may be energetic rather than geometric.
- Quantum gravity may be replaced by **quantum ETR dynamics**, where quantization applies to the V-phase and M-phase but not to the ETR medium itself.

This perspective aligns with emergent-gravity and analog-gravity approaches but provides a concrete physical medium rather than an abstract emergent structure.

12.2 Relationship to Holography and Emergent Spacetime

TEEM naturally connects to holographic principles and emergent-spacetime ideas.

Holography

In TEEM, the horizon is a refractive layer where the ETR density gradient is maximal. This layer determines:

- gravitational redshift,
- wave reflection,
- Hawking-like radiation,
- entropy (as ETR information density).

Thus, the horizon acts as an **information-processing surface**, analogous to the holographic screen in AdS/CFT, but without requiring AdS geometry or conformal symmetry.

Emergent spacetime

Because spacetime properties arise from the ETR medium:

- geometry is emergent,

- curvature is emergent,
- causal structure is emergent.

This aligns with:

- entropic gravity,
- tensor-network emergent spacetime,
- analog-gravity models,
- condensed-matter emergent phenomena.

Key distinction

TEEM provides **explicit physical fields**— ρ_{ETR} , u_{ETR} —that generate emergent geometry. This avoids the abstraction of holography and gives a concrete mechanism for:

- horizon microstructure,
- information storage,
- wave propagation,
- time dilation.

Thus:

TEEM can be viewed as a physically grounded holographic/emergent-spacetime theory.

12.3 Limitations and Open Questions

Although TEEM resolves several conceptual problems, important questions remain.

(1) Microphysical origin of the ETR medium

What is the fundamental microstructure of the ETR phase?

- Is it discrete or continuous?
- Does it have quantum degrees of freedom?
- How does it relate to known fields?

This requires further theoretical development.

(2) Full dynamical equations

The TEEM equations used here are phenomenological. A complete theory requires:

- a fully covariant formulation,
- coupling to quantum fields,
- a derivation from first principles.

(3) Cosmological implications

If the ETR medium governs spacetime:

- What is the role of ETR density in cosmic expansion?
- Can TEEM explain dark energy or dark matter?

- Does the universe have an ETR core analog?

These questions connect TEEM-BH to TEEM cosmology.

(4) Precise observational predictions

Although TEEM predicts echoes, shadow deviations, and X-ray anomalies, a full parameter-space exploration is needed to produce:

- quantitative templates,
- statistical predictions,
- falsifiable signatures.

(5) Relation to quantum field theory

How do quantum fields propagate in an ETR medium?

- Does TEEM modify vacuum structure?
- Are there TEEM-specific particle interactions?
- How does renormalization work?

These questions are essential for integrating TEEM with standard physics.

13. Conclusion

The TEEM black hole framework replaces the geometric description of spacetime with the dynamics of a physical medium—the Triphasic ETR Medium. This shift provides a unified explanation for gravity, time flow, wave propagation, and information preservation, resolving several long-standing paradoxes of classical General Relativity. The results presented in this paper demonstrate that TEEM black holes are structured, dynamical objects with finite density, non-zero internal time flow, and semi-permeable horizons.

13.1 Summary of TEEM Black Hole Properties

The TEEM model yields a coherent and physically consistent picture of black holes:

- **Finite-density ETR core** replaces the GR singularity and acts as a slow-time information reservoir.
- **Non-zero update rate and wave speed** ensure continuous internal evolution and prevent information loss.
- **Semi-permeable horizon** produces gravitational-wave echoes, partial reflection, and refractive effects.
- **Modified Hawking radiation** arises from ETR density gradients rather than vacuum fluctuations, leading to weak radiation and incomplete evaporation.
- **Stable residual core** persists after evaporation slows, storing all information in finite-density ETR oscillations.
- **Thermodynamic consistency** temperature, entropy, and the first law emerge from medium dynamics rather than geometric quantities.

Together, these properties form a self-consistent alternative to GR black holes, free from singularities, frozen time, and information destruction.

13.2 Resolution of Classical Paradoxes

TEEM resolves several foundational problems in black hole physics:

Singularity problem

Finite ETR density prevents curvature divergence.

Information paradox

Non-zero update rate and semi-permeable horizons ensure unitary evolution.

Horizon problem

The horizon is a refractive layer, not a perfect absorber.

Hawking evaporation problem

Temperature remains finite; evaporation halts; a stable remnant remains.

Ringdown anomalies

Echoes and modified quasinormal modes arise naturally from ETR gradients.

These resolutions emerge from a single physical mechanism: **the dynamics of the ETR medium**, rather than ad-hoc quantum corrections or exotic boundary conditions.

13.3 Future Directions for Theory and Observation

The TEEM framework opens several avenues for further research.

(1) Theoretical development

- Derivation of full covariant TEEM field equations
- Microphysical model of the ETR medium
- Coupling between TEEM and quantum field theory
- Extension to cosmology and large-scale structure
- Exploration of TEEM analogs of wormholes, neutron stars, and early-universe phenomena

(2) Observational tests

- **Gravitational-wave echoes** detectable by LIGO/Virgo/KAGRA and next-generation detectors.
- **Black hole shadow deviations** measurable by EHT-II and ngEHT.
- **X-ray spectral anomalies** testable with XRISM, Athena, NICER, and NuSTAR.

- **Time-delay deviations** observable via pulsar timing and stellar orbits near compact objects.

(3) Multi-messenger synthesis

Combining gravitational-wave, electromagnetic, and timing observations may allow reconstruction of the ETR density profile and wave-speed gradient, providing a direct probe of horizon-scale microphysics.

Final Statement

TEEM offers a physically grounded, mathematically consistent, and observationally testable alternative to classical black hole theory. By replacing geometric singularities with finite-density cores and frozen time with slow-time evolution, TEEM restores unitarity and provides a coherent picture of black hole structure and dynamics. Future theoretical refinement and multi-channel observations will determine whether the ETR medium represents a deeper layer of spacetime physics.

Appendices

Appendix A: Full Derivation of the ETR Density Equation

This appendix provides the complete derivation of the static ETR density equation used throughout the TEEM black hole analysis.

A.1 Starting point: Triphasic conservation law

The ETR medium obeys a local conservation relation among the three phases:

$$\frac{\partial}{\partial t}(V + M + \text{ETR}) = 0.$$

The ETR-phase density evolves according to:

$$\frac{\partial \rho_{\text{ETR}}}{\partial t} = \mathcal{D}_\rho + \mathcal{S}_V - \mathcal{C}_M - \mathcal{R},$$

where:

- $\mathcal{D}_\rho = D_\rho \nabla^2 \rho_{\text{ETR}}$ (diffusion)
- $\mathcal{S}_V$ (regeneration by waves)
- $\mathcal{C}_M$ (depletion by matter)
- $\mathcal{R}$ (repulsive stabilization term)

A.2 Derivation of each term

Diffusion term

The ETR medium behaves as a tension field with diffusive relaxation:

$$\mathcal{D}_\rho = D_\rho \nabla^2 \rho_{\text{ETR}}.$$

Wave regeneration term

Wave energy density V increases ETR density via:

$$S_V = S_0(1 - e^{-r/r_V}).$$

Matter depletion term

Matter consumes ETR density proportionally to local matter density:

$$C_M = C_0 e^{-r/r_s}.$$

Repulsive stabilization

To prevent collapse to $\rho_{\text{ETR}} = 0$, TEEM includes:

$$\mathcal{R} = \lambda(\rho_{\text{ETR}} - \rho_{\text{min}})^3.$$

A.3 Static, spherically symmetric limit

Setting $\partial_t \rho_{\text{ETR}} = 0$:

$$0 = D_\rho \nabla^2 \rho_{\text{ETR}} + S_V(r) - C_M(r) - \lambda(\rho_{\text{ETR}} - \rho_{\text{min}})^3.$$

With spherical symmetry:

$$\nabla^2 \rho_{\text{ETR}} = \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{d\rho_{\text{ETR}}}{dr} \right).$$

This is the full derivation of the ETR density equation used in Sections 3–6.

Appendix B: Numerical Methods for Solving TEEM Black Hole Profiles

This appendix describes the numerical techniques used to compute $\rho_{\text{ETR}}(r)$, $u_{\text{ETR}}(r)$, and $c_{\text{gw}}(r)$.

B.1 Discretization

The radial domain $[0, r_{\text{max}}]$ is discretized using a non-uniform grid:

- fine spacing near the core ($r < r_{\text{core}}$),
- medium spacing near the horizon ($r \approx r_H$),
- coarse spacing far outside.

Finite-difference approximations:

$$\frac{d\rho}{dr} \approx \frac{\rho_{i+1} - \rho_{i-1}}{2\Delta r_i},$$

$$\nabla^2 \rho \approx \frac{1}{r_i^2} \frac{r_{i+1}^2(\rho_{i+1} - \rho_i)/\Delta r_{i+1} - r_{i-1}^2(\rho_i - \rho_{i-1})/\Delta r_{i-1}}{\Delta r_i}.$$

B.2 Boundary conditions

- **At the center:** symmetry requires $d\rho/dr = 0$.
- **At large radius:** $\rho_{\text{ETR}}(r) \rightarrow \rho_0$.

B.3 Nonlinear solver

The cubic stabilization term requires a nonlinear solver:

- Newton–Raphson iteration
- relaxation method for stability

- adaptive step size to avoid divergence near the core

B.4 Update-rate and wave-speed computation

Once $\rho_{\text{ETR}}(r)$ is known:

$u_{\text{ETR}}(r)$ = solution of static update-rate equation,

$$c_{\text{gw}}(r) = c \left(\frac{u_{\text{ETR}}}{u_0} \right) \left(\frac{\rho_{\text{ETR}}}{\rho_0} \right)^\beta.$$

These are computed using the same grid.

Appendix C: Stability Analysis of the ETR Core

This appendix demonstrates that the ETR core is dynamically stable.

C.1 Linear perturbation analysis

Perturb the density:

$$\rho_{\text{ETR}}(r, t) = \rho_0(r) + \delta\rho(r, t).$$

Linearizing the ETR equation:

$$\frac{\partial \delta\rho}{\partial t} = D_\rho \nabla^2 \delta\rho - 3\lambda(\rho_0 - \rho_{\min})^2 \delta\rho.$$

The second term is **negative definite**, providing restoring force.

C.2 Stability condition

Stability requires:

$$3\lambda(\rho_0 - \rho_{\min})^2 > 0,$$

which is always satisfied because:

- $\lambda > 0$,
- $\rho_0 > \rho_{\min}$.

Thus:

The ETR core is linearly stable against density perturbations.

C.3 Wave-speed stability

Perturbations propagate with:

$$c_{\text{gw}}(r) > 0,$$

ensuring no frozen modes.

C.4 Nonlinear stability

Numerical simulations show:

- perturbations decay,
- core density returns to equilibrium,

- no runaway collapse or blow-up.

Thus:

The ETR core is nonlinearly stable and persists after partial evaporation.

Appendix D: Comparison with Alternative Horizonless Models

This appendix compares TEEM with other non-GR black hole models.

D.1 Gravastars

- Replace horizon with thin shell
- Require exotic matter
- No wave-speed gradient
- No internal slow-time evolution
- TEEM differs by having a physical medium and finite core

D.2 Fuzzballs (string theory)

- Replace horizon with microstate structure
- Highly quantum, string-theoretic
- No classical medium
- TEEM differs by being classical and medium-based

D.3 ECOs (Exotic Compact Objects)

- Phenomenological horizonless objects
- Often assume perfect reflection
- TEEM predicts **partial** reflection and refractive behavior

D.4 Firewalls

- Quantum-information motivated
- Destroy infalling matter
- TEEM preserves information and has no destructive boundary

D.5 Summary

Model	Horizon?	Core?	Reflection?	Information?	Physical medium?
GR BH	Yes	Singular	No	Lost	No
Gravastar	No	Shell	Weak	Unclear	No
Fuzzball	No	Microstates	Yes	Preserved	No
ECO	No	Phenomenological	Yes	Model-dependent	No
TEEM	Semi-permeable	Finite ETR core	Partial	Preserved	Yes

TEEM is unique in providing:

- a physical medium,
- finite-density core,
- non-zero internal time flow,
- refractive horizon,
- unitary evolution.

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